Dispersion of χ_e, χ_m.

$$\vec{P} = \varepsilon_0 \chi_e \vec{E} \quad , \vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 (1+\chi_e) \vec{E} = \varepsilon \vec{E}$$

$$\vec{M} = \chi_m \vec{H}$$

But in general $\chi_e \neq$ const because response of the medium is <u>not</u> instantaneous.

For dispersive linear medium

$$\vec{P}(t) = \varepsilon_0 \int_{-\infty}^{\infty} \chi_e (t-t') \vec{E}(t') \, dt' \left[= \varepsilon_0 \int G_e(t-t') \vec{E}(t') dt' \right.$$

in the textbook

$\vec{P}$ appears in response to $\vec{E}$ $\Rightarrow$ t must be after t' $\Rightarrow$ $t-t'>0$

<u>Causality</u>: $\chi_e (t-t') = 0$ for $t < t'$ $(t-t'<0)$

$$\vec{P}(t) = \varepsilon_0 \int_{-\infty}^{t} \chi_e(t-t') \vec{E}(t') dt'$$

[Cause ($\vec{E}$) and effect ($\vec{P}$)]

Fourier: $\int dt \, e^{i\omega t} \vec{P}(t) = \varepsilon_0 \int dt \, e^{i\omega t} \int dt' \, \chi_e(t-t') \vec{E}(t')$

let $\chi(\Omega) = \int d\tau \, e^{i\Omega\tau} \chi(\tau) \Rightarrow \chi(\tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \chi(\omega)$

$$\Rightarrow \vec{P}(\omega) = \varepsilon_0 \int dt \, dt' \int \frac{d\Omega}{2\pi} e^{-i\Omega(t-t')} \chi(\Omega) e^{i\omega t} \vec{E}(t')$$

but $\int_{-\infty}^{\infty} dt \, e^{i t(\omega - \Omega)} = 2\pi \delta(\omega - \Omega)$

$$\Rightarrow \vec{P}(\omega) = \varepsilon_0 \int dt' \, e^{i\omega t'} \chi(\omega) \vec{E}(t') = \varepsilon_0 \chi(\omega) \vec{E}(\omega)$$

$\vec{P}$, $\vec{E}$ and $\chi(t)$ are real $\Rightarrow \chi^*(t) = \chi(t)$

$$\Rightarrow \chi(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \chi(\omega)$$

$$\chi(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} \chi(t) \Rightarrow \chi^*(\omega) = \int_{-\infty}^{\infty} dt\, e^{-i\omega t} \chi^*(t)$$

this limit is actually 0 as $\chi(t<0)=0$

$\boxed{\text{for real } \omega}$

$$= \int dt\, e^{i(-\omega)t} \chi(t)$$

$$= \chi(-\omega).$$

More generally, consider complex $\underline{\omega}$

$$\cancel{\chi^*(t) = \int \frac{d\omega^*}{2\pi}}$$

$$\cancel{\chi^*(\omega^*) = \int dt\, e^{-i\omega^* t} \chi^*(t) =}$$

$$\cancel{= \int dt\, e^{i\omega^* t} \chi(t) = \chi(-\omega^*)}$$

Thus, $\chi(\omega) = \chi_1(\omega) + i\chi_2(\omega) = Re[\chi] + i\, Im[\chi]$

$$\chi^*(\omega) = \chi_1(\omega) - i\chi_2(\omega)$$

$$\chi(-\omega) = \chi_1(-\omega) + i\chi_2(-\omega)$$

$$\Rightarrow \underbrace{\chi_1(\omega) = \chi_1(-\omega)}_{\text{even}}, \quad \underbrace{\chi_2(\omega) = -\chi_2(-\omega)}_{\text{odd}}.$$

$\chi_e(t) = 0$ for $t < 0$

To account for causality, we require that

$$\vec{E}(t') \longrightarrow \underbrace{\vec{E}(t')}_{\text{regular function}} \underbrace{e^{\delta t'}}_{\text{adiabatic factor}} \quad \text{with } \delta = +0 \text{ so that } e^{\delta t'} \xrightarrow[t' \to -\infty]{} 0$$

Thus, in the far past $(t' = -\infty)$ external perturbation $\vec{E}(t')$ is absent.

$$\vec{P}(t) = \varepsilon_0 \int dt' \, \chi_e(t-t') \, \vec{E}(t') \, e^{\delta t'} =$$

$$\left(e^{\delta t'} = e^{(t'-t)\delta} e^{+\delta t} \right) =$$

$$= \varepsilon_0 \, e^{\delta t} \int dt' \, \chi_e(t-t') \, e^{-\delta(t-t')} \, \vec{E}(t')$$

Observe that $\vec{P}(t) \sim e^{\delta t}$ so that $\vec{P}(t \to -\infty) \to 0$

Importantly, $\chi(t-t') \to \chi(t-t') e^{-\delta(t-t')}$

$$\Rightarrow \chi(\omega) = \int_{-\infty}^{\infty} dt \, e^{i\omega t} \chi(t) e^{-\delta t} =$$

$$= \int_{-\infty}^{\infty} dt \, e^{i(\omega + i\delta)t} \chi(t) = \int_{0}^{\infty} dt \, e^{i(\omega + i\delta)t} \chi(t)$$

Causality requirement produces infinitesimal imaginary part in ω: $\omega \to \omega + i\delta$, with $\delta = +0$.

because $\chi(t) = 0$ for $t < 0$.

Thus $\varepsilon(\omega) = \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{i(\omega + i\delta)\tau} \chi(\tau) \right] =$ (more generally)

$$= \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{i\omega\tau} \chi(\tau) \right]$$

where $\omega = \omega' + i\omega'' = \omega_1 + i\omega_2$ is complex variable now

For $\omega'' > 0$ $e^{i\omega\tau} = e^{i\omega_1\tau} e^{-\omega_2\tau} \xrightarrow[\tau \to \infty]{} 0$

$\Rightarrow$ the integral $\int_0^\infty d\tau \, e^{i\omega\tau} \chi(\tau)$ is certain to converge

$\Rightarrow$ $\varepsilon(\omega)$ has no singularities in upper complex plane $\omega' + i\omega''$, $\omega'' > 0$.

This does not apply to lower plane, where $\omega'' < 0$!
(so that $e^{-\omega''\tau} = e^{|\omega''|\tau}$ is not bounded)

$$\underset{\hookrightarrow \omega_1 + i\omega_2}{\varepsilon(\omega)} = \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{i\omega_1\tau - \omega_2\tau} \chi(\tau) \right]$$

$$\underset{\hookrightarrow \omega_1 - i\omega_2}{\varepsilon(\omega^*)} = \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{i\omega_1\tau + \omega_2\tau} \chi(\tau) \right]$$

$$\varepsilon(-\omega^*) = \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{-i\omega_1\tau - \omega_2\tau} \chi(\tau) \right]$$

$-\omega^* = -\omega_1 + i\omega_2$

but $\varepsilon^*(\omega) = \varepsilon_0 \left[1 + \int_0^\infty d\tau \, e^{-i\omega_1\tau - \omega_2\tau} \chi(\tau) \right]$

$\hookrightarrow$ because $\chi^*(\tau) = \chi(\tau)$

$\Rightarrow$ $\varepsilon(-\omega^*) = \varepsilon^*(\omega)$

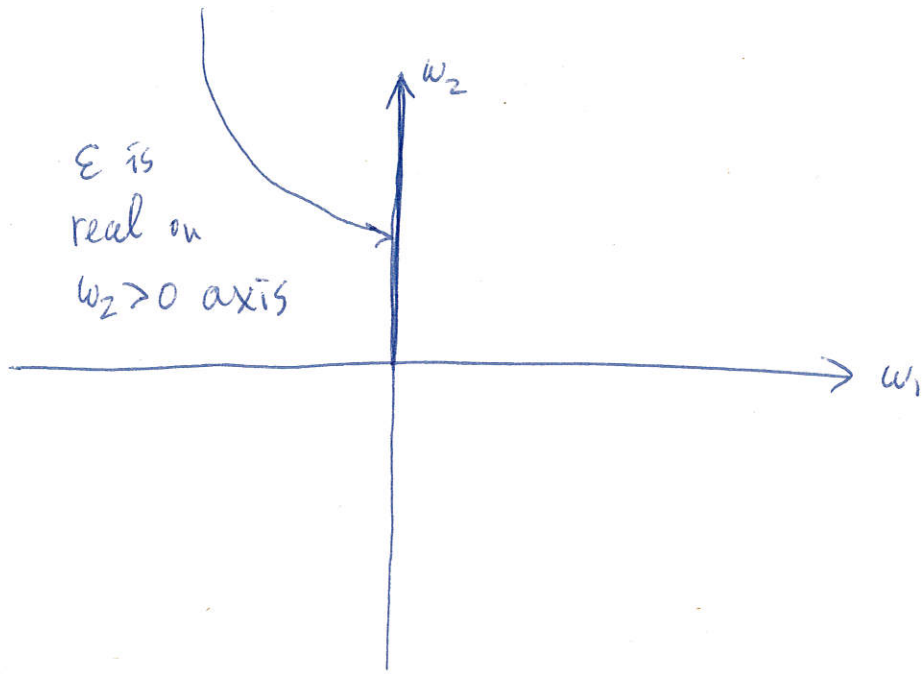
$\chi(-\omega^*) = \chi^*(\omega)$

here $\omega = \omega' + i\omega''$ is complex!

$\Rightarrow$ for example, for purely imaginary $\omega = i\omega_2$ $(\omega_1 = 0)$

$$-\omega^* = i\omega_2$$

$$\Rightarrow \quad \varepsilon(i\omega_2) = \varepsilon^*(i\omega_2)$$



$$\vec{D}(\omega) = \varepsilon(\omega)\,\vec{E}(\omega)$$

For $\vec{E}(t) = \vec{E}_0 e^{-i\omega t} + \vec{E}_0^* e^{i\omega^* t}$

$$\cancel{\vec{E}(\omega) = \int dt\, e^{i\omega t}\vec{E}(t) = \vec{E}_0\, 2\pi\delta(\omega-\omega)}$$

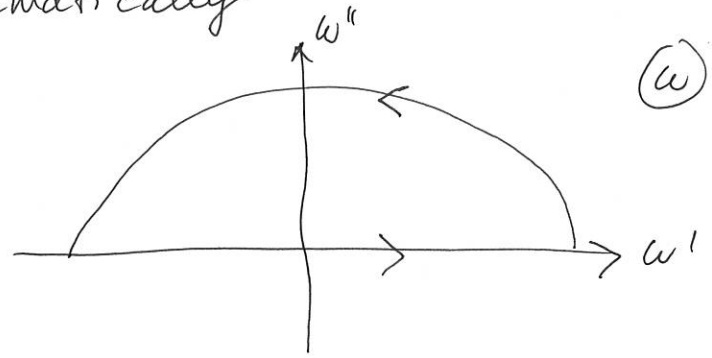
For real $\omega = \omega^*$, we get again $\chi^*(\omega) = \chi(-\omega)$.

Kramers-Kronig: $\tau = t - t'$

$$\varepsilon(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega\tau} \varepsilon(\omega)$$

For $\tau < 0$ $(t < t')$ $\varepsilon(\tau) = 0$

Mathematically



$$e^{-i(\omega' + i\omega'')\tau} = e^{-i\omega'\tau} e^{\omega''\tau} \to 0$$

for $\omega'' \to \infty, \tau < 0$.

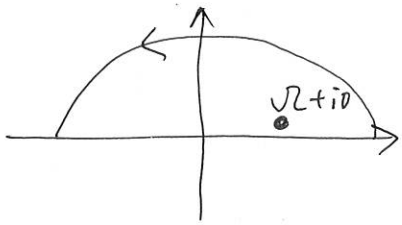
we get $\varepsilon(t<0) = 0$ if $\varepsilon(\omega)$, as a function of complex $\omega = \omega' + i\omega''$, has <u>no poles</u> in <u>$\omega'' > 0$ half-plane</u>.

Consider now $f(\omega) = \dfrac{\varepsilon(\omega) - \varepsilon_0}{\omega - \Omega - i0}$

(Note $\varepsilon(\omega) - \varepsilon_0 = \varepsilon_0 \chi(\omega)$ which converges at $\omega \to \infty$)

Ω = real frequency

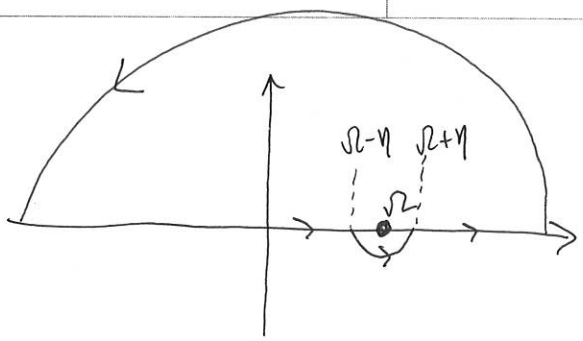
$$I = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi i} f(\omega) = \oint \frac{d\omega}{2\pi i} f(\omega) = \frac{2\pi i}{2\pi i} \operatorname{Res}\left(\frac{\varepsilon(\omega) - \varepsilon_0}{\omega - (\Omega + i0)}\right) =$$



$$= \varepsilon(\Omega) - \varepsilon_0$$

since $\varepsilon(\omega)$ does not have any poles for $\omega'' > 0$!

On the other hand



$$I = \underbrace{\int_{-\infty}^{\Omega-\eta} \frac{d\omega}{2\pi i} f(\omega) + \int_{\Omega+\eta}^{\infty} \frac{d\omega}{2\pi i} f(\omega)}_{\text{Principal part } \frac{1}{2\pi i} P\int_{-\infty}^{\infty} d\omega \frac{\varepsilon(\omega)-\varepsilon_0}{\omega-\Omega}} + \underbrace{\int_{\curvearrowright} \frac{d\omega}{2\pi i} \frac{\varepsilon(\omega)-\varepsilon_0}{\omega-\Omega}}_{I_2}$$

$$I_2 : \quad z = \omega - \Omega = \eta e^{i\phi}$$

$$I_2 = \frac{\varepsilon(\Omega)-\varepsilon_0}{2\pi i} \int_C \frac{dz}{z} \quad , \quad \int_C \frac{dz}{z} = \int \frac{d(e^{i\varphi})}{e^{i\varphi}} = i\int_{\pi}^{2\pi} d\varphi = i\pi$$

$$\Rightarrow I = \frac{1}{2\pi i} P\int_{-\infty}^{\infty} d\omega \frac{\varepsilon(\omega)-\varepsilon_0}{\omega-\Omega} + \frac{i\pi}{2\pi i}\left(\varepsilon(\Omega)-\varepsilon_0\right)$$

But $I = \varepsilon(\Omega) - \varepsilon_0$

$$\Rightarrow \quad \varepsilon(\Omega) - \varepsilon_0 = \frac{1}{\pi i} P\int_{-\infty}^{\infty} d\omega \frac{\varepsilon(\omega)-\varepsilon_0}{\omega-\Omega}$$

So what we derived is well-known Sokhotski identity:

$$\frac{1}{\omega-\Omega-i0} = P\frac{1}{\omega-\Omega} + i\pi\,\delta(\omega-\Omega)$$

$$\frac{1}{\omega-\Omega+i0} = P\frac{1}{\omega-\Omega} - i\pi\,\delta(\omega-\Omega).$$

$$\Rightarrow \varepsilon'(\Omega) - \varepsilon_0 + i\,\varepsilon''(\Omega) = -\frac{i}{\pi}\,P\int_{-\infty}^{\infty} d\omega\,\frac{\varepsilon'(\omega)-\varepsilon_0+i\varepsilon''(\omega)}{\omega-\Omega}$$

Kramers-Kronig

$$\begin{cases} \varepsilon'(\Omega) - \varepsilon_0 = \dfrac{1}{\pi}\,P\displaystyle\int_{-\infty}^{\infty} d\omega\,\frac{\varepsilon''(\omega)}{\omega-\Omega} \\[2ex] \varepsilon''(\Omega) = -\dfrac{1}{\pi}\,P\displaystyle\int_{-\infty}^{\infty} d\omega\,\frac{\varepsilon'(\omega)-\varepsilon_0}{\omega-\Omega} \end{cases}$$

<u>Conductors</u> $\varepsilon(\omega) = \varepsilon_{reg}(\omega) + \dfrac{i\sigma_0}{\omega}$

↳ regular function, no poles

↳ pole at $\omega=0$

Kramers–Kronig relations hold for $\underline{\varepsilon_{reg}(\omega) = \varepsilon(\omega) - \dfrac{i\sigma_0}{\omega}}$

$$\Rightarrow \begin{cases} \varepsilon'(\Omega) - \varepsilon_0 = P\displaystyle\int_{-\infty}^{\infty}\frac{d\omega}{\pi}\,\frac{\varepsilon''(\omega)-\sigma_0/\omega}{\omega-\Omega} \\[2ex] \varepsilon''(\Omega) = \dfrac{\sigma_0}{\Omega} - P\displaystyle\int_{-\infty}^{\infty}\frac{d\omega}{\pi}\,\frac{\varepsilon'(\omega)-\varepsilon_0}{\omega-\Omega} \end{cases}$$

(Note that $P\displaystyle\int_{-\infty}^{\infty}\frac{d\omega}{\pi}\,\frac{\sigma_0}{\omega(\omega-\Omega)} = 0$.)

$$P\int \frac{d\omega}{\omega(\omega-\Omega)} = P\int d\omega \left(\frac{1}{\omega} - \frac{1}{\omega-\Omega}\right)\frac{1}{-\Omega}$$

But $P\int d\omega \frac{1}{\omega-\Omega} = \lim_{\eta\to 0}\left\{\int_{-\infty}^{\Omega-\eta} \frac{d\omega}{\omega-\Omega} + \int_{\Omega+\eta}^{\infty} \frac{d\omega}{\omega-\Omega}\right\} =$

$$= \lim_{\eta\to 0}\left\{\int_{-\infty}^{-\eta} \frac{dx}{x} + \int_{\eta}^{\infty} \frac{dx}{x}\right\} = 0$$

$$x = -y \qquad \int_{\infty}^{\eta} \frac{-dy}{-y} = \int_{\infty}^{\eta} \frac{dy}{y} = -\int_{\eta}^{\infty} \frac{dx}{x}$$

$$\Rightarrow \quad P\int \frac{d\omega}{\omega(\omega-\Omega)} = 0 \quad \text{indeed.}$$

$\Rightarrow$ <u>for conductors</u>

$$\boxed{\begin{aligned} \varepsilon'(\Omega) - \varepsilon_0 &= P\int_{-\infty}^{\infty} \frac{d\omega}{\pi} \frac{\varepsilon''(\omega)}{\omega-\Omega} \\ \varepsilon''(\Omega) &= \frac{\sigma_0}{\Omega} - P\int_{-\infty}^{\infty} \frac{d\omega}{\pi} \frac{\varepsilon'(\omega)-\varepsilon_0}{\omega-\Omega} \end{aligned}}$$

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$, $\vec{D} = \varepsilon_0 (1+\chi_e) \vec{E} = \varepsilon \vec{E}$, $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$\vec{M} = \chi_m \vec{H}$

Induced current $\vec{j}_b = \frac{\partial \vec{P}}{\partial t} + \vec{\nabla} \times \vec{M}$

$\boxed{\text{Alternative formulation}}$: <u>generalized displacement</u> $\vec{\mathcal{D}}$

let $\underset{(\vec{r},t)}{\vec{\mathcal{D}}} = \varepsilon_0 \underset{(\vec{r},t)}{\vec{E}} + \int_{-\infty}^{t} dt' \vec{j}_b(\vec{r},t') = \varepsilon_0 \vec{E} + \vec{\mathcal{P}}(\vec{r},t)$

where <u>generalized polarization</u> $\vec{\mathcal{P}} = \int_{-\infty}^{t} dt' \vec{j}_b(r,t')$

Then $\vec{j}_b(r,t) = \frac{\partial \vec{\mathcal{P}}}{\partial t}$ and $\rho_b(r,t) = -\vec{\nabla} \cdot \vec{\mathcal{P}}$.

since $\vec{\nabla} \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \frac{\partial}{\partial t}\left(\vec{\nabla} \cdot \vec{\mathcal{P}} + \rho\right) = 0$

Then Maxwell's equs read:

$$\vec{\nabla} \times \vec{E} + \frac{\partial}{\partial t} \vec{B} = 0 \qquad (1)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \qquad (2)$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

$\hookrightarrow \vec{J}_{ext} + \vec{J}_b$

$$= \mu_0 \vec{J}_{ext} + \mu_0 \left(\varepsilon_0 \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{\mathcal{P}}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \vec{j}_{ext} + \mu_0 \frac{\partial \vec{\mathcal{D}}}{\partial t} \qquad (3)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho \Rightarrow \vec{\nabla} \cdot \left(\varepsilon_0 \vec{E}\right) = \rho_{ext} + \left(-\vec{\nabla} \cdot \vec{\mathcal{P}}\right)$$

$$\Rightarrow \vec{\nabla} \cdot \vec{\mathcal{D}} = \rho_{ext} \qquad (4)$$

And so we have 4 equs and 3 fields: $\vec{B}$, $\vec{E}$ and $\vec{\mathcal{D}}$.

Need to add constitutive relations which we write as

$$\vec{\mathcal{D}} = \tilde{\varepsilon}\, \vec{E}$$

(Note that in general $\varepsilon \neq \tilde{\varepsilon}$ but they are closely related.) Also note that this formulation does not require field $\vec{H}$! Everything is contained in $\vec{\mathcal{D}}$.

See discussion in Ch. 6, pp. 64-74, for more arguments. [Melrose, McPhedran book]

They write $\boxed{\text{eq. 6.15}}$ $(J_{ind})_i(\omega, \vec{k}) = \sigma_{ij}(\omega, \vec{k})\, E_j(\omega, \vec{k})$

where $\vec{J}_{ind}$ is the induced current $= \vec{J}_b$ in our notations.

From $\vec{\mathcal{D}} = \tilde{\varepsilon}\, \vec{E} = \varepsilon_0 \vec{E} + \vec{\mathcal{P}}$ we get

$$\vec{\mathcal{P}} = (\tilde{\varepsilon} - \varepsilon_0)\, \vec{E} \quad \text{and} \quad \vec{j}_b = \vec{j}_{ind} = \frac{\partial}{\partial t} \vec{\mathcal{P}}$$

$$\Rightarrow \quad \vec{j}_{ind}(r,t) = \frac{\partial}{\partial t}(\tilde{\varepsilon} - \tilde{\varepsilon}_0)\, \vec{E}(r,t)$$

↳ this in general implies non-local in space-time relation.

Upon Fourier transform we get

$$\left(\vec{j}_{ind}\right)_i(\omega, \vec{k}) = -i\omega\left(\tilde{\varepsilon}_{ij} - \varepsilon_0 \delta_{ij}\right) E_j(\omega, \vec{k})$$

$$\Rightarrow \quad \sigma_{ij}(\omega, k) = -i\omega\left(\tilde{\varepsilon}_{ij} - \varepsilon_0 \delta_{ij}\right)$$

$$\tilde{\varepsilon}_{ij}(\omega,\vec{k}) = \varepsilon_0 \delta_{ij} + \frac{\sigma_{ij}(\omega,\vec{k})}{-i\omega} = \varepsilon_0 \delta_{ij} + \frac{i\,\sigma_{ij}(\omega,\vec{k})}{\omega}$$

$$= \varepsilon_0 \left(\delta_{ij} + \frac{i\,\sigma_{ij}(\omega,\vec{k})}{\varepsilon_0 \omega} \right) = \varepsilon_0 K_{ij}(\omega,\vec{k})$$

$\boxed{\text{eq. 6.17}}$